(10)

Complexity to be reduced.

(i) $\overset{a}{23}|\overset{b}{4}$ (or) (ii) $\overset{a}{2}|\overset{b}{34}$

Step2:- In the above 2 cases knowing 23^2 is easy than 34^2 so, first one is preferable. If you know both squares then both are preferable

Note:- Here we must know 5 digited highest perfect square i.e., $316^2 = 99856$

means squares of 100 to 316 all are resultant to 5 digited numbers only.

↳ 6 digited least perfect square is $317^2 = 100489$.

↳ 6 digited highest perfect square is $999^2 = 998001$

↳ 7 digited least perfect square is $1000^2 = 1000000$

↳ Conclusions:-

Resultant digits of squares from 1 to 3 → 1 digit

" " " " 4 to 9 → 2 digits

" " " " 10 to 31 → 3 digits

" " " " 32 to 99 → 4 digits

" " " " 100 to 316 → 5 digits

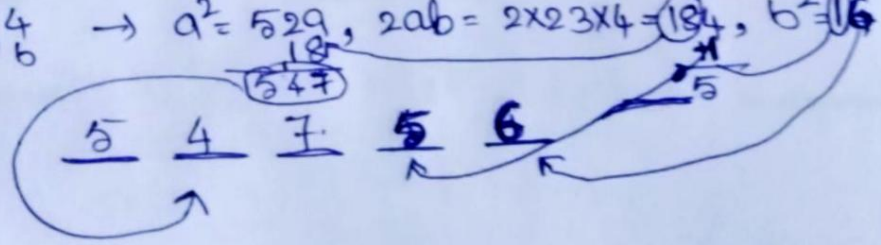
" " " " 317 to 999 → 6 digits

" " " " 1000 to 3162 → 7 digits

[∵ 7 digited highest perfect square is $3162^2 = 9998244$]

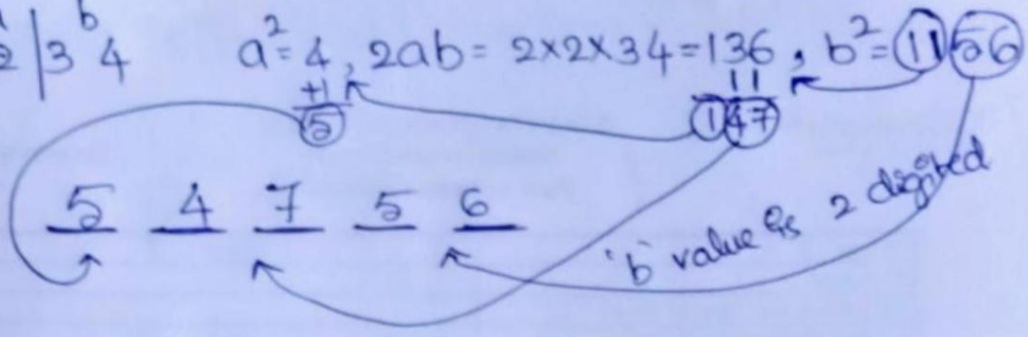
Step3:- Place a^2, $2ab$ & b^2 values into resultant spaces according to no. of digits in 'b'.

(i) $\overset{a}{23}|\overset{b}{4}$ → $a^2 = 529$, $2ab = 2 \times 23 \times 4 = 184$, $b^2 = 16$



(11)

(vii) $\overset{a}{2}\,|\,\overset{b}{3}\,4$ $\quad a^2=4,\ 2ab=2\times2\times34=136,\ b^2=1156$



(viii) Product of two 2 digited numbers when

① Unit digits sum is '10' ② 10's positions are same

Ex:- 1 :- $78 \times 72 \longrightarrow x\,a \times x\,b$

$x(x+1) \mid a \times b$

$\underset{(7\times(7+1))}{5\ 6}\ \underset{(8\times2)}{1\ 6}$

Ex:- 2 :- $13\underset{x=21}{3} \times 13\underline{7}$ [Other than units positions are same]

$\underset{(13\times(13+1))}{1\ 8\ 2\ 2\ 1}$ (7×3)

(ix) Product of 2 two digited numbers :-

① Simple :- Ex:- 31×21 $\quad [a_1=3, b_1=1;\ a_2=2, b_2=1]$

assume $a_1b_1 \times a_2b_2$

do → $a_1a_2 \mid a_1b_2+a_2b_1 \mid b_1b_2$

$3\times2 \mid 3\times1+1\times2 \mid 1\times1$

$6 \mid 5 \mid 1$

$3\ 1 \times 2\ 1$ (a_1 b_1 a_2 b_2)

Resultant is 3 digited number so you can place 6,5 & 1 for the 3 digits as it is $\Rightarrow \underline{6}\ \underline{5}\ \underline{1}$

② Complex :- Ex:- 86×49 (a_1b_1 a_2b_2)

$a_1a_2 \mid a_1b_2+a_2b_1 \mid b_1b_2$

$8\times6 \mid 8\times9+4\times6 \mid 6\times9$

$32 \mid 72+24 \mid 54$

$= 96$